MANIPAL INSTITUTE OF TECHNOLOGY
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**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING**

**III SEMESTER B.TECH.**

**MIDTERM**

SUBJECT: COMPUTER ORGANIZATION AND ARCHITECTURE (CSE 2121)

Time: 10:45 AM – 12:15 PM Date: 23/09/2024 MAX.MARKS: 30

| Q.No. | Question | M |
|---|---|---|
| 11 | With the help of flow chart, perform unsigned multiplication between two binary numbers where multiplicand = 101101 and multiplier =10001. Show all the steps clearly | 4 |

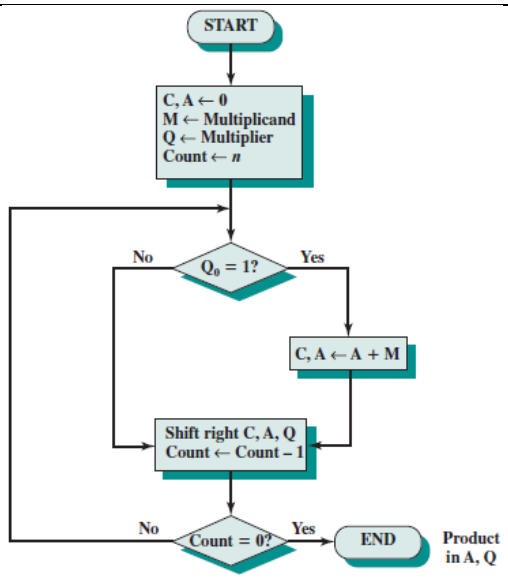



Figure 10.9 Flowchart for Unsigned Binary Multiplication

1M

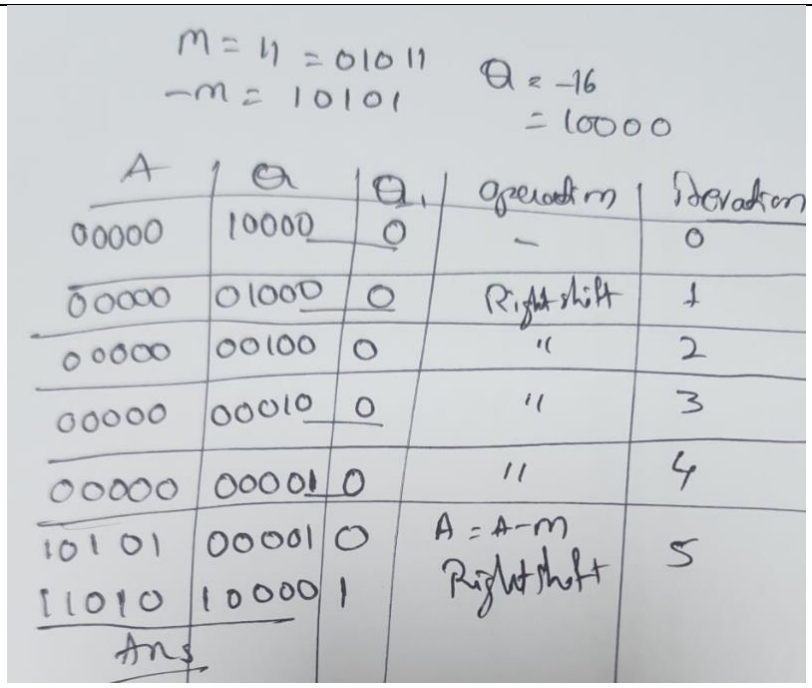
(Q) multiplier = 10001 $(17)_{10}$ multiplicand = 101101 (M) $(45)_{10}$ $(17 \times 45) = (765)_{10}$

| C | A | Q | M | |
|---|---|---|---|---|
| 0 | 000000 | 010001 | 101101 | Initial |
| | 101101 | | | (A+M) ① |
| 0 | 101101 | | | |
| 0 | 010110 | 101000 | | (shift) |
| 0 | 001011 | 010100 | | (shift) ② |
| 0 | 000101 | 101010 | | shift ③ |
| 0 | 000010 | 110101 | | shift ④ |
| | 101101 | | | (A+M) |
| 0 | 101111 | 110101 | | |
| 0 | 010111 | 111010 | | (shift) ⑤ |
| 0 | 001011 | 111101 | | shift ⑥ |

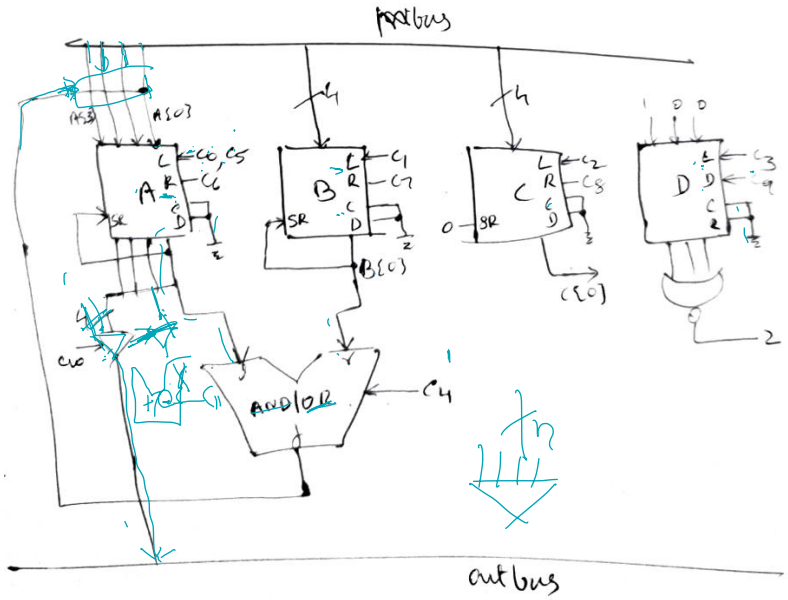
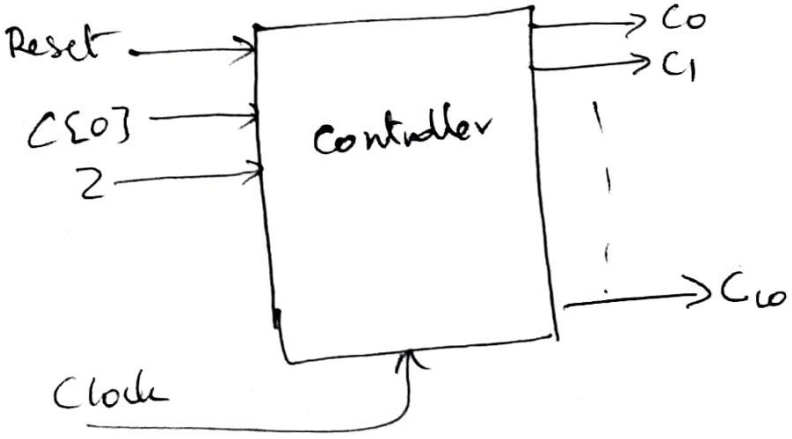
product = A & Q = $(001011\ 111101)_2 = (765)_{10}$

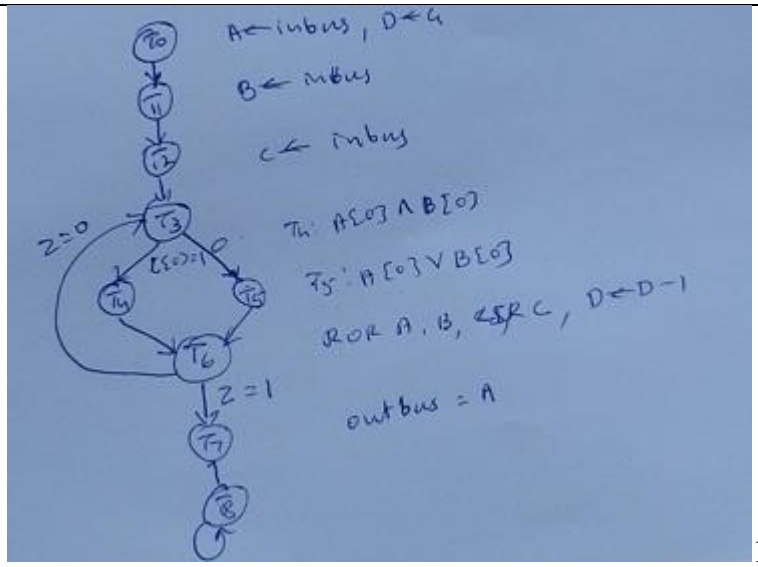
3M

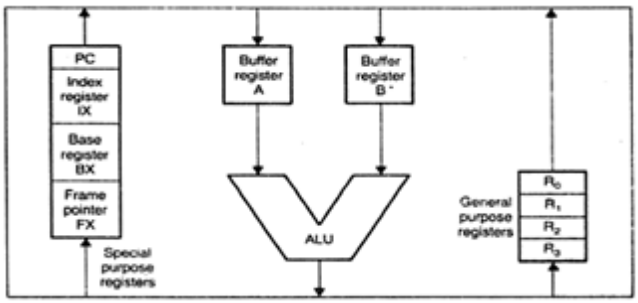
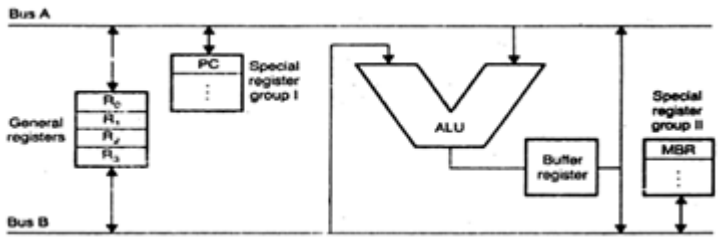
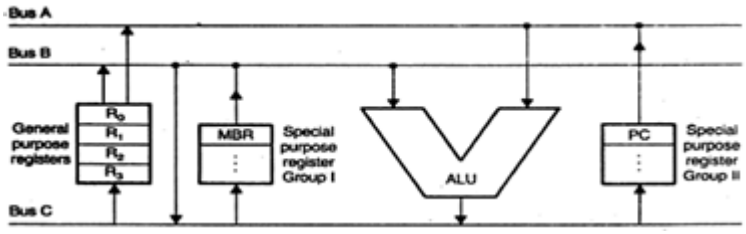
| | | |
|---|---|---|
| 12 | Show the detailed steps in multiplying the following two numbers using Booth's multiplication algorithm: M= +11 and Q= -16.<br>Scheme:<br>Correct number of iterations or exact required number of bits=0.5M<br>Correct Answer: 0.5M<br>Other correct steps: 0.5*4= 2M<br>Answer: | 3 |

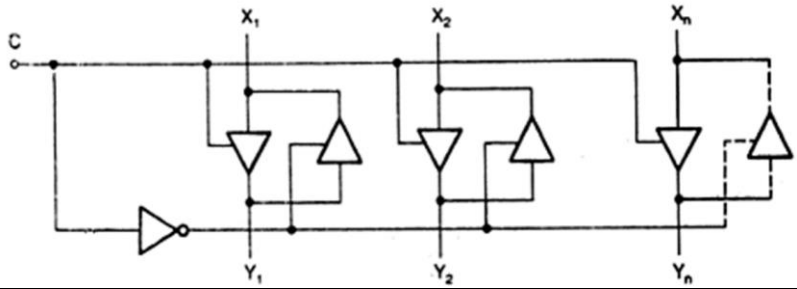



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|---|---|---|
| 13 | Design a processing unit establishing the required control points for the Register Transfer Description (RTD) shown in Fig. P13.<br>Declare registers A[4], B[4], C[4], D[3];<br>Declare inbus[4], outbus[4];<br>START: A ← inbus, D ← 4;<br>B ← inbus;<br>C ← inbus;<br>BACK: if C[0]=1 then A[0]←A[0]Λ B[0];<br>go to ROTATE;<br>A[0]←A[0] V B[0];<br>ROTATE: ROR (A), ROR(B), LSR(C), D← D-1;<br>if D>0 then go to BACK;<br>outbus=A;<br>HALT: go to HALT;<br>Fig. P13<br>**Answer:**<br>C0: A←inbus<br>C3: d←4<br>C1: B←inbus<br>C2: C←inbus<br>C4: f←l Λ r C4': f←l V r 1.5M<br>C5: A[0] = f<br>C6: ROR(A)<br>C7: ROR(B)<br>C8: LSR(C)<br>C9: D←D-1<br>C10: Outbus=A | 3 |

| | | |
|---|---|---|
| |  <br>1.5M | |
| 14 | Design the block diagram of the controller and develop the corresponding state diagram for the RTD given in Fig. P13.<br>**Answer:**<br>  <br>T0: C0, C3<br>T1: C1<br>T2: C2<br>T3: none<br>T4: C4, C5<br>T5: C5<br>T6: C6, C7, C8, C9<br>T7: C10<br>T8: none 1.5M | 3 |

| | | |
|---|---|---|
| |  1.5M | |
| 15 | Calculate the sum of two positive binary numbers 1110.1 and 10111.1 in IEEE 32-bit representation. Show all the steps clearly and represent the final answer in IEEE 32-bit format.<br>**Answer:**<br>**Step i:** Normalize **(0.5)**<br>1110.1→ After normalizing: $1.1101 \times 2^3$<br>E'=3+127= 130= $10000010_{(2)}$<br>In IEEE 32-bit format: 0 10000010 11010000000000.....<br><br>**Step ii:** Normalize **(0.5)**<br>10111.1→ After normalizing: $1.01111 \times 2^4$<br>E'=4+127= 131= $10000011_{(2)}$<br>In IEEE 32-bit format: 0 10000011 011110000000000.....<br><br>**Step 1 :** **(0.25)**<br>Choose the number with the smaller exponent and shift its mantissa right a number of steps equal to the difference in exponents.<br>14.5 is with smaller mantissa, hence: 1. 1101000...→0.11101000…<br><br>**Step 2:** **(0.25)**<br>Set the exponent of the result equal to the larger exponent<br>10000011 (exponent of 23.5)<br><br>**Step 3:** **(0.5)**<br>Perform addition on the mantissas and determine the sign of the result<br>1.011110000000000 +<br>0.111010000000000<br>10.011000000000000<br>**Step 4:** **(0.5)**<br>Normalize the resulting value: $1.0011000... \times 2^1$<br>E'= 10000011+00000001=$10000100_{(2)}$<br><br>**In 32-bit format:** 0 10000100 001100000000 **(0.5)** | 3 |

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|---|---|---|
| 16 | Elaborate the single, two and three bus orientation for the CPU to perform R2 ← R0 + R1.<br>Single BUS: Consider operation R2 ← R0 + R1 (0.5M)<br>• 1st CLK : Tx R0 contents to A reg<br>• 2nd CLK : Tx R1 contents to B reg<br>• 3rd CLK: ALU produces SUM and result is loaded to R2<br>• ➔ 3 CLK cycles are needed here to complete R2 ← R0 + R1 operation<br>  <br><br>(0.5M)<br>Two BUS: (0.5M)<br>▪ 1st CLK cycle: loading operands and storing result in O/P buffer<br>▪ 2nd CLK cycle: Addition operation & result pushed to buffer reg. The result then is put R2 either via BUS A or BUS B.<br>  <br><br>(0.5M)<br>Three Bus: (0.5M)<br>• This can be performed 1 one CLK<br>  <br><br>(0.5M) | 3 |
| 17 | With a neat diagram, design and explain bi-directional data bus.<br>Scheme: 1 mark for correct diagram | 2 |

| | | |
|---|---|---|
| | 1 Mark for the logic and explanation on working of the same<br>Answer:<br>When C=1, X to Y<br><br>When C=0, Y to X<br><br>  <br> | |
| 18 | Consider a floating-point system that uses 8-bit numbers: 3 bits for the exponent (with bias = 3) and 4 bits for the mantissa (excluding the implied leading 1 for normalized numbers). Illustrate the rounding method to represent the binary number 101.0011 in this floating-point system.<br>**Answer:**<br><br>$101.0011_2 = 1.010011X2^2$ after normalising<br>Any $0.b_{-1}b_{-2}b_{-3}b_{-4}1$ . . is rounded to $0.b_{-1}b_{-2}b_{-3}\ b_{-4} + 0.0001$,<br>Hence, 1.010011 becomes 1.0101 | 2 |
| 19 | Calculate the sum of two signed 8-bit binary numbers A= 11011001 and B=10110110 given in the two's complement representation. Is there an overflow in this addition? Justify your answer.<br>**Answer:**<br><br>11011001<br>10110110<br>10001111<br><br>Since the carry-in and carry-out in MSB are the same there is no overflow in this example. | 2 |